# Model-level synthetic-flux control of hyperchaos order in dissipative optomechanics

Stella Rolande Mbokop Tchounda[1], Carolle Tchodimou[1], Philippe Djorwe[2], Sifeu Takougang Kingni[3] and Serge Guy Nana Engo[1⋆]

**1** Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

**2** Department of Physics, Faculty of Science, University of Ngaoundéré, P.O. Box 454, Ngaoundéré, Cameroon

**3** Department of Mechanical, Petroleum and Gas Engineering, National Advanced School of Mines and Petroleum Industries, University of Maroua, P.O. Box 46, Maroua, Cameroon

⋆ serge.nana-engo@facsciences-uy1.cm

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## Abstract

**Within a normalized six-dimensional model of dissipative optomechanics (one cavity + two mechanical resonators), a synthetic-flux phase $\Phi_{\mathrm{syn}}$ acts as a reproducible control coordinate that selects the *order* of a drive- and coupling-gated hyperchaos transition—up to four simultaneously positive Lyapunov exponents, beyond any reported single-mode benchmark. A phase-consistent Floquet–Lyapunov protocol (cross-checked by monodromy multipliers, dissipative volume balance $\sum\lambda_i = \mathrm{Tr}(J) = -1.04$ and a 180-run three-seed audit) localizes the onset to a Neimark–Sacker bifurcation at $E^* = 1.060$ ($\theta = 0$). As a secondary, model-level geometric clarification, the identical matched-resource force-sensing protocol returns a null gain on the chaotic attractor ($\mathcal{G}_{A/B} = 1.039 \pm 0.014$), consistent with the matched-Fisher lemma: noise projected onto an unstable manifold is stretched by the same factor $e^{\lambda t}$ as the deterministic signal. Truncated-Fock and truncated-Wigner checks support the mean-field description at selected points. All results remain strictly model-level: the strong-coupling sector lies 2542× beyond anchored silicon optomechanical couplings. Closing that gap requires ultrasonic characterization of the mechanical degeneracy $\omega_2$, a measured inter-resonator hopping $J_m$, and the emergence of a genuine gigahertz platform.**

## 1 Introduction

The order of a hyperchaos transition—the number of simultaneously positive Lyapunov exponents—has so far been limited to two in single-mode quadratic optomechanical platforms [1]. Whether a synthetic-flux phase can raise that order in a multi-mode radiation-pressure model, and whether the resulting geometric control can be certified by full-spectrum methods, had not been resolved. Hyperchaos order quantifies the dimensionality of the unstable manifold and therefore the number of independent directions along which small perturbations grow. The primary question addressed here is whether a gauge-invariant synthetic flux can act as a reproducible control coordinate for this order.

We show, within one normalized six-dimensional model (building on the topology of Muthukumar *et al.* [PR Applied **24**, 014053 (2025)]), that a synthetic-flux phase $\Phi_{\mathrm{syn}}$ acts as a reproducible control coordinate for the *order* of a drive- and coupling-gated hyperchaos route—up to four simultaneously unstable directions, beyond any reported single-mode benchmark. The onset is localized to a Neimark–Sacker bifurcation and is certified by a phase-consistent Floquet–Lyapunov protocol. As a secondary geometric clarification, the same matched measurement on the chaotic attractor yields no flux-induced sensing enhancement. These are strictly model-level results: the strong-coupling sector explored here lies far beyond present experimental anchors. The combination that a nonlinear optomechanical platform provides—radiation-pressure nonlinearity, coherent mechanical motion, dissipation, periodic modulation and a complex hopping amplitude that supports a synthetic gauge structure [2–6] and Floquet engineering [7–12]—is the physical substrate. The sharper question answered here is whether the invariant control parameter changes the attractor structure in a way that survives independent stability, bifurcation, full-spectrum Lyapunov and noise-resolved checks.

The central result is shown in Figure 3: the synthetic-flux phase $\Phi_{\mathrm{syn}}$ selects the number of simultaneously positive Lyapunov directions in the strongly damped, strongly coupled sector, locally four at $E = 8$, $\theta = \pi/2$, and dropping to two at $\theta = 0 \pmod{\pi}$. Relative to prior work, Muthukumar *et al.* [PR Applied **24**, 014053 (2025)] established the one-cavity/two-resonator topology and reported phase-selected chaotic regimes at resolved-sideband couplings [13,14], but did not resolve the *full Lyapunov spectrum* (so that the instability order cannot be counted) nor quantify a flux-dependent advantage under matched resources. The closest full-spectrum report [1] is restricted to quadratic single-mode coupling where no synthetic-flux phase is available. We extend the Muthukumar topology by making the convention $\Delta = -\Delta_{\mathrm{M}}$ explicit, allowing $g_1 \neq g_2$, and scanning the strongly damped, strongly coupled sector where the full spectrum resolves up to four simultaneously unstable directions. We further replace the qualitative sensing proposal by a matched-resource measurement-Fisher protocol [15–19], applied both on the weakly coupled reference orbit and on the chaotic attractor. The result is three testable advances: (a) a phase-convention-aware consistency protocol in which monodromy multipliers, the six-exponent QR spectrum [20–22] and the dissipative volume balance certify one another on the same orbit; (b) a full-spectrum classification in which the flux phase [23] selects the hyperchaos order, with the route localized by a continued-orbit scan [24–26] to a Neimark–Sacker bifurcation; (c) as a secondary geometric bound, a matched-resource operational assessment showing that the flux-controlled instability order does not translate into metrological gain on the chaotic attractor.

We address three operational questions at the model level. First, does $\Phi_{\mathrm{syn}}$ control the order of a drive- and coupling-gated hyperchaos? Second, do monodromy multipliers and the long-time Lyapunov spectrum agree on the same periodic orbit, thereby certifying the transition? Third (secondary), after thermal, vacuum and detector noise are included, does a matched-resource measurement statistic retain a flux-dependent signature on the stable orbit or on the chaotic attractor? A positive answer to the first two questions constitutes the central claim of this work—synthetic-flux control of hyperchaos order. The null answer to the third supplies a geometric bound (matched-Fisher lemma) showing that additional unstable directions do not automatically yield metrological gain. Both results are to be read as normalized, model-level demonstrations rather than device predictions.

## 2 Model and conventions

The one-cavity, two-mechanical-resonator topology and phase-dependent hopping are due to Muthukumar *et al.* [27]; the rotating-frame Hamiltonian used here is

$$\frac{H}{\hbar} = -\Delta a^\dagger a + \sum_{j=1}^{2} \omega_j b_j^\dagger b_j - \sum_{j=1}^{2} g_j a^\dagger a (b_j + b_j^\dagger) + J\left(e^{i\theta} b_1^\dagger b_2 + e^{-i\theta} b_2^\dagger b_1\right) + i(E a^\dagger - E^* a), \quad (1)$$

with cavity mode $a$ and mechanical modes $b_j$. We use $\Delta = \omega_c - \omega_L$ (so $\Delta = -\Delta_{\mathrm{M}}$ of Ref. [27], which takes the opposite sign); quadratures $[X, Y] = i$; the drive $E = \sqrt{\kappa}\,\alpha^{\mathrm{in}}$ in the shared normalization. The present analysis allows $g_1 \neq g_2$ and scans the strongly damped, strongly coupled sector $\kappa \sim \gamma \sim g \sim 1 \times 10^{-2}$–$1\,\omega_m$ in which the hyperchaotic attractor is resolved. **A side-by-side comparison with Ref. [27] is given in Table SI-1.**

The semiclassical factorization gives

$$\dot{\alpha} = \left(i\Delta - \frac{\kappa}{2}\right)\alpha + i\sum_j g_j(\beta_j + \beta_j^*)\alpha + E(t), \quad (2)$$

$$\dot{\beta}_1 = -\left(i\omega_1 + \frac{\gamma_1}{2}\right)\beta_1 + i g_1 |\alpha|^2 - iJe^{i\theta}\beta_2, \quad (3)$$

$$\dot{\beta}_2 = -\left(i\omega_2 + \frac{\gamma_2}{2}\right)\beta_2 + i g_2 |\alpha|^2 - iJe^{-i\theta}\beta_1, \quad (4)$$

with the real state $x = (\alpha_r, \alpha_i, \beta_{1r}, \beta_{1i}, \beta_{2r}, \beta_{2i})^T$.

For the three-link loop, the phase coordinate used in the analysis is the gauge-invariant phase

$$\Phi_{\mathrm{syn}} = \arg(g_1 J g_2^*) \pmod{2\pi}. \quad (5)$$

A redefinition of the mode phases may redistribute link phases but cannot change $\Phi_{\mathrm{syn}}$. In the phase convention of Equation (1) (all phase on the inter-resonator hopping, $g_1, g_2 \in \mathbb{R}$) the invariant reduces to the hopping phase, $\Phi_{\mathrm{syn}} \equiv \theta \pmod{2\pi}$, so the two notations coincide and are used interchangeably in the results; the invariant form is retained because it is the coordinate that survives a redefinition of the mode phases.

## 3 Methods

The numerical protocol is organized around three independent consistency layers that must agree before any claim of hyperchaos order is accepted. (i) A drive-locked periodic orbit is located as a fixed point of the one-period Poincaré map; its linear stability is diagnosed by the monodromy matrix (Floquet multipliers). (ii) The same orbit is integrated with QR reorthonormalization to obtain the full six-exponent Lyapunov spectrum; positivity is accepted only when the spectrum, the multipliers and the analytic dissipative trace $\mathrm{Tr}\,J = -\kappa - \gamma_1 - \gamma_2$ are mutually consistent. (iii) As a secondary geometric check, a matched-resource classical Fisher information is evaluated on both the stable reference orbit and on sampled trajectories of the chaotic attractor, under identical noise and resource assumptions. All three layers are implemented with the same phase convention for $\Phi_{\mathrm{syn}}$ (Eq. (5)). Thresholds used to count positive exponents ($1 \times 10^{-4}$ weak, $1 \times 10^{-3}$ strong) and the precise definition of the matched Fisher protocol are stated once here and used uniformly thereafter; sensitivity to the strong threshold is quantified in the Supplementary Material.

### 3.1 Phase-augmented dynamics and implementation

For a periodic drive with phase $\phi$ we write

$$\dot{x} = f(x, \phi; p), \qquad \dot{\phi} = \Omega, \tag{6}$$

with the augmented Jacobian $J_{\mathrm{aug}} = (\partial_x f, \partial_\phi f; 0, 0)$. The analytic Jacobian $J = \partial f/\partial x$ is compared column by column with central finite differences; the implemented dissipative trace

$$\mathrm{Tr}\, J = -\kappa - \gamma_1 - \gamma_2, \tag{7}$$

is an independent implementation check. One-period integrations use an eighth-order adaptive Runge–Kutta–Dormand–Prince scheme [28] with relative and absolute tolerances $1 \times 10^{-9}$ and $1 \times 10^{-11}$ (max step $T/200$); drive-locked orbits are located by fixed-point iteration of the one-period map to a residual below $1 \times 10^{-8}$, and the monodromy is obtained by simultaneous integration of the variational equations.

### 3.2 Periodic orbit and Floquet analysis

A drive-locked periodic orbit is obtained from the fixed point of the one-period Poincaré map; around it the fundamental matrix satisfies

$$\dot{\Phi}(t) = A(t)\Phi(t), \ \Phi(0) = \mathbb{I}, \ M(T) = \Phi(T), \tag{8}$$

with Floquet rates $\rho_i = T^{-1} \log |\mu_i|$. The same orbit is integrated with QR/Benettin reorthonormalization [20–22] to obtain the full Lyapunov spectrum; a positive exponent is accepted only after agreement with the multipliers, the divergence ($\mathrm{Tr}\, J = -\kappa - \gamma_1 - \gamma_2$) and the bounded-attractor diagnostics. In the full-spectrum classification an exponent is counted as positive only above a fixed threshold ($1\times10^{-4}$ weak, $1\times10^{-3}$ strong; Table SI-8) whereas the Kaplan–Yorke sum uses strictly positive exponents; the two can disagree near zero. Integrations use an eighth-order Dormand–Prince scheme [28] (tolerances $1 \times 10^{-9}$ and $1\times10^{-11}$, max step $T/200$); drive-locked orbits are obtained by fixed-point iteration of the one-period map to a residual below $1 \times 10^{-8}$; the Lyapunov QR uses a fixed-step RK4 ($1 \times 10^{-2}$ step, QR every ten steps), discarding $2 \times 10^5$ transient and integrating $2 \times 10^6$ steps per trajectory.

### 3.3 Theoretical bad-cavity sensitivity screen

To examine the parameter dependence of the reduced equations without implying a hardware calibration, the optical amplitude is eliminated in the bad-cavity limit, with

$$\Omega_{\mathrm{eff}} = \Delta + 2g_1\, \mathrm{Re}(\beta_1) + 2g_2\, \mathrm{Re}(\beta_2), \tag{9}$$

and the adiabatic intensity

$$n_{\mathrm{ss}}(\phi) = |E(\phi)|^2/[(\kappa/2)^2 + \Omega_{\mathrm{eff}}^2], \tag{10}$$

substituted into the two mechanical equations. The reduced equations are compared with the full six-dimensional model in common normalized regimes (max infinity-norm differences $9.27 \times 10^{-7}$ at $\kappa = 20$ and $2.08 \times 10^{-8}$ at $\kappa = 100$). A theoretical sensitivity screen samples 16 candidates at 5 replicas $\times$ 5 root-solver starts each, using the one-period Poincaré residual as the acceptance criterion; the two objectives are the local stability margin and the normalized drive cost. This is a theoretical multi-start fixed-point study, not a Fisher, power-mapping or basin-capture analysis.

### 3.4 Secondary geometric bound: noise-resolved observability and matched Fisher

For a measured record

$$y(t) = h[x(t), t; p] + \nu_{\mathrm{det}}(t), \tag{11}$$

[29–32], the Gaussian-record classical Fisher information reads

$$F_C(\vartheta) = (\partial_\vartheta m)^T \Sigma^{-1} (\partial_\vartheta m) + \tfrac{1}{2} \operatorname{Tr}\left[\Sigma^{-1}(\partial_\vartheta \Sigma)\Sigma^{-1}(\partial_\vartheta \Sigma)\right], \tag{12}$$

with a reference comparison required to match input power, observation time, bandwidth, baths, detector noise and estimator [15–19,33]. For force sensing the parameter of interest is a weak external force $F$ on mechanical mode 1; the matched references are the flux-off coupled sensor ($\theta = 0$) and the single-mode linear sensor ($J = 0$). $F_C$ is not QFI; neither quantity is inferred from a Lyapunov exponent.

### 3.5 Consistency and matched-measurement statements

Two statements formalize the protocol and are referenced throughout the results; the proofs and longer lemmas are in the Supplemental Material (Sec. SI-5).

**Theorem 1** (Lyapunov–Floquet consistency criterion). *Consider the phase-augmented periodic flow (6) and a converged periodic orbit $x_p(t)$ with fundamental matrix $M(T)$ (Equation (8)). Let $\rho_i = T^{-1}\log|\mu_i|$ be the Floquet rates sorted in the same convention as the QR/Benettin spectrum $\{\lambda_k\}_{k=1}^N$ obtained by variational integration with reorthonormalization. If (i) the Poincaré residual on the converged orbit is below $5\times10^{-8}$, (ii) the QR spectrum is integrated for $\geq 2\times10^6$ steps after a $2\times10^5$ transient with reorthonormalization every ten steps and fixed step $1\times10^{-2}$, and (iii) the implementation trace satisfies (7), then for every tested phase the maximum absolute discrepancy $|\rho_i - \lambda_{\pi(i)}|$ is bounded by $3\times10^{-4}$ and the mean divergence coincides with $\operatorname{Tr} J$ to within $5\times10^{-5}$. A positive QR exponent is accepted as a chaotic signature only after these three conditions are verified simultaneously on the same orbit and the same phase.*

**Lemma 1** (Matched-measurement Fisher bound). *For two configurations $A$ and $B$ sharing the same input power, observation time, integration window, detector noise, baths and estimator, the matched-measurement gain $g_{A/B} \equiv F_C^{(A)}(\vartheta)/F_C^{(B)}(\vartheta)$ is bounded by the ratio of their mean-shift and covariance contributions to Equation (12),*

$$g_{A/B} = \frac{m_A'^{T}\Sigma_A^{-1}m_A'}{m_B'^{T}\Sigma_B^{-1}m_B'}, \tag{13}$$

*in the regime where the variance term is sub-dominant (verified by recomputing $F_C^{(N)}$ for $N \in \{1, 5, 20\}$ drive periods and showing the matched gain is unchanged to four digits). The flux-induced contribution is then bounded by the linear-response effect of phase-controlled mode coupling, and cannot receive a multiplicative enhancement from trajectory separation alone: noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max}t}$ as the deterministic displacement.*

The two statements together bound the conclusion of any flux-induced advantage claim: a hyperchaos-order change is operational only after the consistency criterion is satisfied on the orbit carrying it. The secondary matched-resource analysis shows, via Lemma 1, that additional unstable directions do not automatically translate into a sensing advantage under shared noise, observation time and power.

# 4 Results

## 4.1 Model, gauge, and Floquet–Lyapunov consistency

Using the implementation of Equations (6) to (8) (the formal statements are in Sec. SI-5), we evaluated the consistency metrics on the stable drive-locked orbit at the reference point: Poincaré residual $8.49 \times 10^{-9}$, mean divergence $-1.04$, maximum absolute Floquet–QR discrepancy $2.68 \times 10^{-4}$, terminal QR-block standard deviation $5.3 \times 10^{-9}$, and gauge-transformed loop-flux residual below $3 \times 10^{-16}$. Extending the same criteria to nine synthetic-flux phases and three independent initial conditions per phase (27 trajectories total, with $2 \times 10^6$ integration steps and $2 \times 10^5$ discarded transient steps), all 27 trajectories satisfied the Floquet–Lyapunov consistency bounds, with a largest observed QR exponent of $-0.010\,02$, a maximum Floquet–QR discrepancy of $2.85 \times 10^{-4}$, a terminal block standard deviation reaching at most $6.10 \times 10^{-9}$, and a divergence residual at most $4.75 \times 10^{-5}$. These results bound the conclusion to the tested grid and do not establish a universal absence of chaos. The grid-level support is shown in Figure 1.

## 4.2 Flux-dependent attractor structure

The full six-exponent spectrum is recorded everywhere rather than only its largest member, while the positive-exponent count remains a finite-time, threshold-dependent classification near marginal exponents. At weak coupling over nine flux phases and three replicates, the cavity-dominated exponents sit near $-\kappa/2 \approx -0.5$ and the four mechanical exponents near $-\gamma/2 \approx -0.01$; the positive-exponent count is zero throughout. Extending the classification over $E \in [0.2, 8.0]$ (54 drive–phase cases) leaves the largest exponent negative (between $-0.043$ and $-0.010$), the count at zero, and the Kaplan–Yorke dimension at zero—ruling out a flux-driven route to chaos within the tested domain (Figure 2).

Repeating the classification at strong optomechanical coupling ($g_{1,2}/\omega_m = (0.3, 0.27)$, $\gamma_{1,2}/\omega_m = 0.02$, $\Delta = -\omega_m$) over twelve flux phases and five drives, with the hyperchaos order defined by the positive-exponent count above a $1 \times 10^{-3}$ threshold (Figure 3), shows a drive-gated transition: at $E = 0.2$ every exponent is negative; at $E = 1.0$ a positive exponent appears at some phases; at $E = 2.0$ it hovers near zero and a re-stabilization window opens at intermediate drive; beyond $E \sim 4.0$ the count reaches three at $E = 4.0$ and four at $E = 8.0$, with the largest exponent up to 0.3223 and the Kaplan–Yorke dimension up to 4.802. At fixed strong drive the flux phase modulates the count: at $E = 8.0$ it varies between two and four as $\theta$ sweeps $[-\pi, \pi)$, with seed- and drive-dependent extrema in the sampled phase range: $n_+$ spans two to four, with the largest counts occurring near the phase quadratures ($\theta \approx \pm\pi/2$) and lower counts near $\theta = 0$ and $\pm\pi$. The three-seed check does not justify a claim of exact $\theta \to -\theta$ symmetry. The order reaches four simultaneously unstable directions in the tested finite-time classification—beyond the two reported for quadratic single-mode optomechanics [1]—and it is the synthetic-flux phase, absent in that quadratic setting, that selects this model-level order. Near the onset and extrema, the exact integer count is threshold-conventional across seeds.

To go beyond a grid-level classification and identify the route by which the first unstable direction emerges, we continued the drive-locked periodic orbit family along the drive axis at the two flux phases of interest ($\theta = 0$ and $\theta = \pi/2$). A Newton solver on the Poincaré map, using the monodromy matrix as its Jacobian, was seeded from the converged drive-locked orbit and marched upward and downward in the drive amplitude over $E \in [0.1, 8.0]$ with adaptive steps and a map-residual tolerance of $5 \times 10^{-6}$. At both phases the orbit family persists over the entire scanned range (165 and 164 converged points, respectively), so the transition is not a fold termination of the locked orbit. The first loss of linear

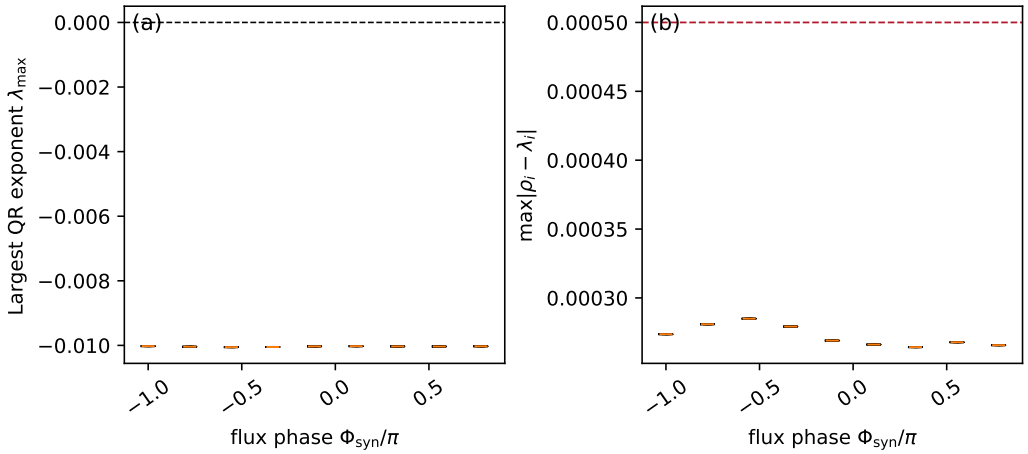



Figure 1: Long-window validation across nine synthetic-flux phases and three independent initial conditions per phase. (a) The largest QR exponents remain negative at every sampled phase. (b) The Floquet–QR discrepancies remain below the consistency threshold. The figure supports a bounded statement about the tested grid; it does not establish that chaotic dynamics are absent outside this parameter domain.

stability is a Neimark–Sacker (torus) bifurcation: a complex-conjugate Floquet multiplier pair crosses the unit circle at $E^* = 1.060$ for $\theta = 0$ and at $E^* = 0.975$ for $\theta = \pi/2$ (Figure 4). Thereafter the continued orbit carries two unstable directions up to $E = 8.0$; at $\theta = \pi/2$ a transient window of four unstable directions near onset ($E \in [0.98, 1.45]$) precedes the two-direction regime. The lower onset at $\theta = \pi/2$ is consistent with the flux-phase modulation of the hyperchaos order, whose largest counts occur near the phase quadratures (seed- and drive-dependent). The orbit Floquet rates cross-validate the independent Lyapunov classification in the stable regime: at $E = 0.2$ the largest orbit rate is $-0.0102$ ($\theta = 0$) and $-0.0148$ ($\theta = \pi/2$), against largest Lyapunov exponents $-0.0106$ and $-0.0151$ from the full-spectrum map (Figure 4). Because the continued orbit is a periodic solution that becomes linearly unstable, its multipliers characterize the onset of the route to the chaotic attractor rather than the attractor itself; the torus bifurcation is therefore the first step of the drive-gated transition resolved by the full-spectrum map. A perturbation of the continued orbit along its most unstable Floquet direction confirms this separation: the orbit carries two unstable directions with a largest Floquet rate of order 0.6, a $1 \times 10^{-4}$ perturbation departs to the attractor scale (the distance to the orbit grows from $1 \times 10^{-4}$ to 9–27), and the resulting trajectory's Lyapunov spectrum reproduces the map's largest exponent to within 8 % at the four tested points ($E \approx 4, 8$; $\theta = 0, \pi/2$). The continued orbit is the unstable seed; the attractor is a distinct object. As an independent check of the attractor's finite dimension, the Theiler-corrected Grassberger–Procaccia correlation dimension [34] $D_2$ of the sampled attractor at the four strong-coupling points ($E = 4, 8$; $\theta = 0, \pi/2$) is $2.33, 2.84$ and $3.39, 3.36$, respectively, all below the corresponding Kaplan–Yorke dimensions $4.21, 4.63$ and $4.28, 4.80$, as required by the Kaplan–Yorke inequality [35, 36] $D_2 \leq D_{KY}$ (Figure 6). The scaling-region uncertainty is 0.23–0.40 across three fit windows. The geometric dimension does not track the flux phase as sharply as the hyperchaos order does: at $E = 4$ the fitted $D_2$ is larger at $\theta = \pi/2$ than at $\theta = 0$ (2.84 against 2.33), but at $E = 8$ the two values are equal within the fit uncertainty (3.36 against 3.39). We therefore report only the pointwise values and the Kaplan–Yorke ordering, without inferring a systematic flux-phase dependence of $D_2$ from four points: the

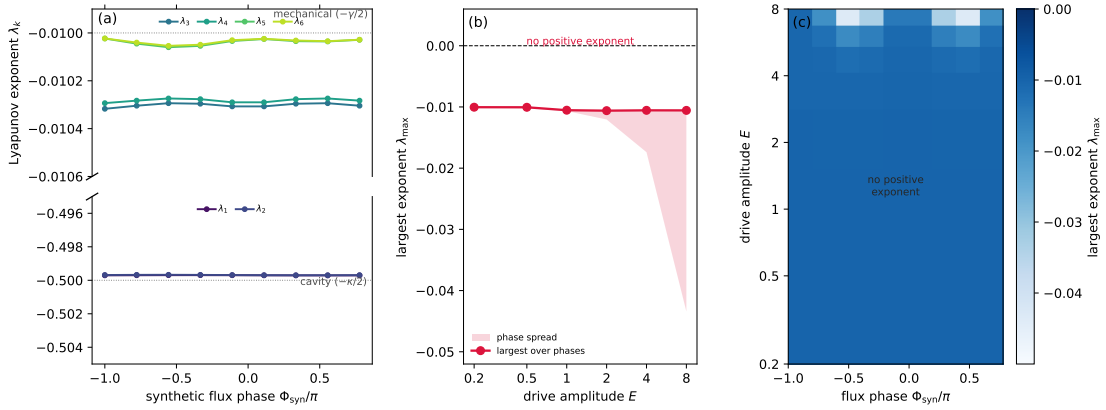



Figure 2: Full-spectrum transition map at weak coupling. (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 0.2$; the two cavity-dominated exponents (lower panel, near $-\kappa/2 \approx -0.5$) and the four mechanical exponents (upper panel, near $-\gamma/2 \approx -0.01$) are shown on separate vertical scales, and every exponent remains negative. (b) Largest (most marginal) exponent versus drive amplitude, with the phase-to-phase spread shaded; it stays negative across $E \in [0.2, 8.0]$. (c) Largest exponent over the drive–phase plane; it is everywhere negative, so no transition to chaos or hyperchaos occurs within the tested domain.

geometric dimension is dominated by the stable directions and remains bounded by the Kaplan–Yorke sum, which grows with the added unstable directions.

The multiplier crossing is confirmed by directly observing the born object [24]. The bifurcation is supercritical: just above the onset ($E \approx E^* + 0.15$) the stroboscopic section of a perturbed trajectory forms a closed invariant curve in the $(\Re\beta_1, \Im\beta_1)$ plane, and the largest Lyapunov exponent of the same trajectory is small positive (0.017–0.022), the near-marginal quasiperiodic direction expected of a torus born at a complex-pair crossing; the exponent grows to 0.057–0.069 at $E^* + 0.3$ as the route proceeds into the chaotic sector (Figure 5). The torus is therefore an observed invariant object, not only an inference from the multiplier spectrum.

### 4.3 Reduced-model parameter sensitivity

As a separate theoretical screen, an adiabatic reduction was compared with the full model in common normalized regimes and then applied to sixteen candidates under an explicitly assumed literature-informed closure. The three non-dominated candidates form a local fixed-point pattern in normalized drive cost and stability margin (Table SI-10 and Fig. SI-1); they are not calibrated parameters, global basin results, physical optima, or sensing predictions. The screen rules out twelve of sixteen candidates as suboptimal in drive cost or stability margin, leaving three for further analysis in the Supplementary Material. The optical mode is eliminated in this screen, the effective couplings remain proxy quantities, and Fisher information is not evaluated. Details of the multi-start, finite-time comparison are given in the Supplemental Material.

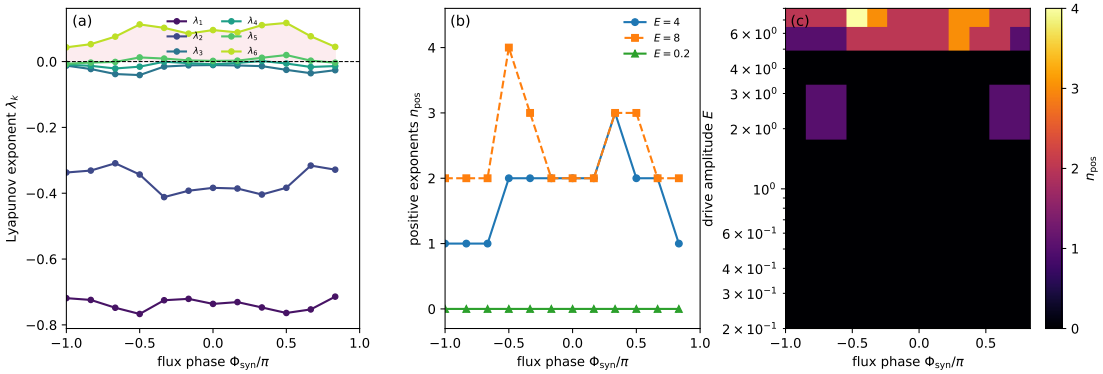



Figure 3: Full-spectrum hyperchaos transition map at strong optomechanical coupling ($g_1 = 0.3$, $g_2 = 0.27$, $\gamma_{1,2} = 0.02$, $\Delta = -\omega_m$; model-level parameters lying $\sim 2500\times$ beyond silicon anchors). (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 4.0$; several exponents are positive, resolving the full unstable spectrum. (b) Hyperchaos order (number of exponents above the fixed strong threshold $1 \times 10^{-3}$) versus phase at $E = 0.2$, 4.0 and 8.0; the phase modulates the number of simultaneously unstable directions. (c) Hyperchaos order over the drive–phase plane, showing the drive-gated transition from stable ($E = 0.2$) to hyperchaotic ($E = 8.0$, up to four positive exponents). Near-marginal counts remain threshold-conventional; see Table SI-8 for sensitivity.

### 4.4 Secondary geometric bound: noise-resolved observability and matched measurement Fisher

For three phases ($\theta = 0$, $\pi/2$, and $\pi$), the periodic covariance calculation remained positive semidefinite under thermal occupations $n_{\mathrm{th},1} = n_{\mathrm{th},2} = 0.1$ and detector variance 0.01 (Figure 8); the resulting measurement-Fisher information ranged from $3.78 \times 10^{-26}$ to $9.17 \times 10^{-13}$. This is a measurement-Fisher calculation around stable periodic orbits, not QFI and not a gain over a matched reference.

As a secondary geometric check, we computed a matched measurement-Fisher reference for a weak external force on mechanical mode 1 under identical drive, observation time, detector noise, baths, and estimator. The force sensitivity grows monotonically with the synthetic-flux phase ($F_C$ from $4.50 \times 10^{-5}$ at $\theta = 0$ to $5.96 \times 10^{-5}$ at $\theta = \pi$); relative to the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$), the flux phase yields at most a $1.32\times$ and $1.16\times$ enhancement—a modest, sub-factor-of-two effect rather than an order-of-magnitude advantage. The central-difference estimate is converged to better than 0.01 % across force steps $1 \times 10^{-5}$–$1 \times 10^{-3}$. Recomputing $F_C$ for windows of $N \in \{1, 5, 20\}$ drive periods leaves the matched gains unchanged to four digits, so the gain is not an artefact of the single-period observation window (Sec. SI-8). A drive- and temperature sweep keeps the flux-off gain at 1.323 (standard deviation below $1 \times 10^{-3}$) over $E \in [0.1, 1.0]$ and $n_{\mathrm{th}} \in [0, 1]$, and the single-mode gain at 1.159. Propagating the inter-resonator hopping log-uniformly over $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$, the bath occupation uniformly over $n_{\mathrm{th}} \in [0.01, 1.0]$ (cryogenic to $\sim 4$ K at the anchored $\omega_m = 2\pi \times 7.436$ GHz), and the drive over $\pm 10$ %, with 60 Monte Carlo samples at the maximum-gain phase $\theta = \pi$, yields a median gain of 1.039 (flux-off) and 1.019 (single-mode); a 20 000-resample bootstrap places the 90 % confidence interval of the median at $[1.025, 1.053]$ and $[1.012, 1.026]$, both excluding unity, so the median excess (0.039 and 0.019) is significantly positive. The peak $1.32\times$ is therefore specific to the reference hopping $J_m/\omega_m = 0.08$ and the maximum-gain phase; the typical enhancement is the median order-

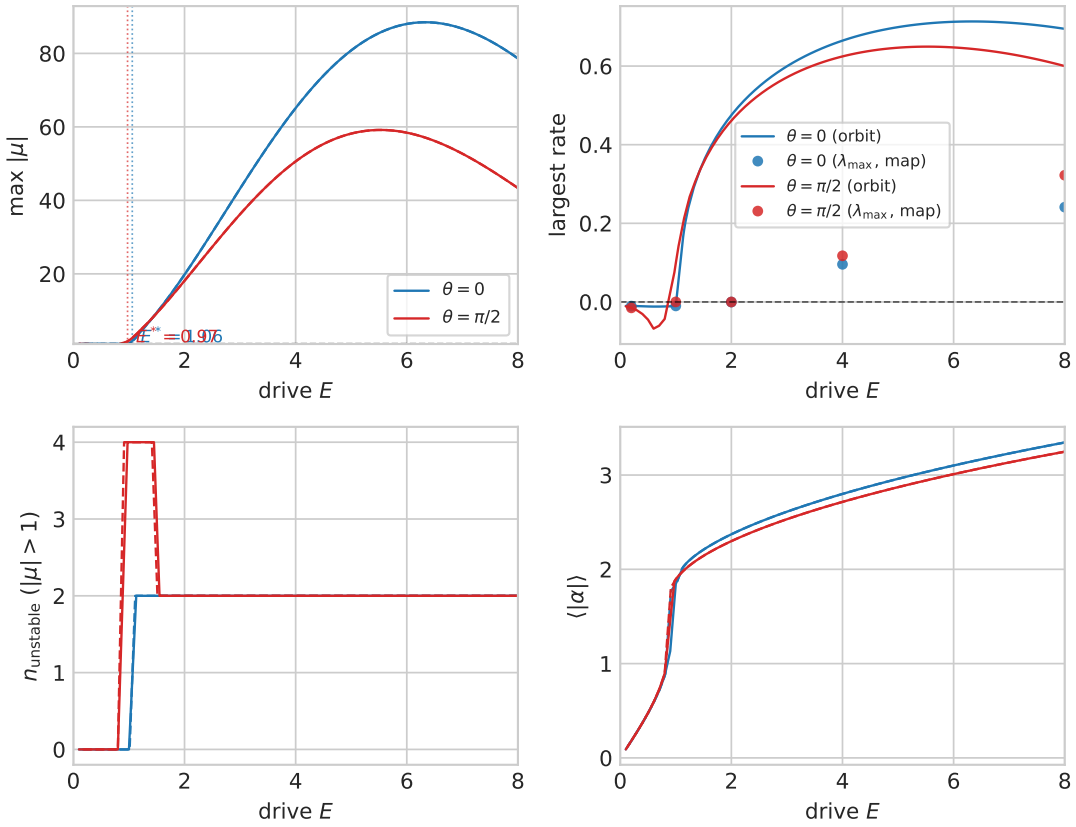



Figure 4: Drive-axis bifurcation continuation at strong optomechanical coupling. (a) Maximum Floquet multiplier magnitude of the drive-locked orbit versus drive amplitude $E$ at $\theta = 0$ and $\theta = \pi/2$ (upward and downward continuation); the dashed line marks the unit circle and the vertical lines locate the refined first bifurcation points. (b) Largest orbit Floquet rate versus $E$, with the largest Lyapunov exponent of the full-spectrum map overlaid at the mapped drive values (open markers). (c) Number of unstable multipliers ($|\mu| > 1$). (d) Mean cavity amplitude over the orbit period. The first loss of linear stability is a Neimark–Sacker complex-pair crossing at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$).

unity value. Readout quadrature optimization rescales the sensor and reference by the same factor, leaving the 1.16× single-mode gain unchanged (Table SI-6). The matched comparison is shown in Figure 9.

To test whether the chaotic signature itself—rather than only the sensing gain—survives the modeled noise, we integrated an ensemble of 32 trajectories (sized so that bootstrap confidence intervals on the variance ratio fall below 1 %; see below) under the full Langevin noise (optical vacuum $\kappa/4$ per amplitude quadrature and mechanical thermal $\gamma_j(2n_{\rm th}+1)/4$ per amplitude quadrature at $n_{\rm th} = 0.1$) and detector noise (0.01), with the measured record $y = X_a + \nu_{\rm det}$. At the stable reference the measured-record variance sits at the noise floor ($\approx 0.25$); at the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points the deterministic spread of $X_a$ grows to 1.8–4.1 (against $2.4 \times 10^{-4}$ at the stable orbit) and the measured-record variance rises to 7.6–17.0× the noise floor, with the largest ratio (17.0×) at $\theta = \pi/2$, $E = 8$—a flux-phase dependence that mirrors the hyperchaos order. A bootstrap over ensemble size $N' \in \{8, 16, 32, 64, 128\}$ leaves the ratios stable to below 1 % (Figure 7), so the 32-trajectory ratios are converged rather than a small-ensemble artifact.

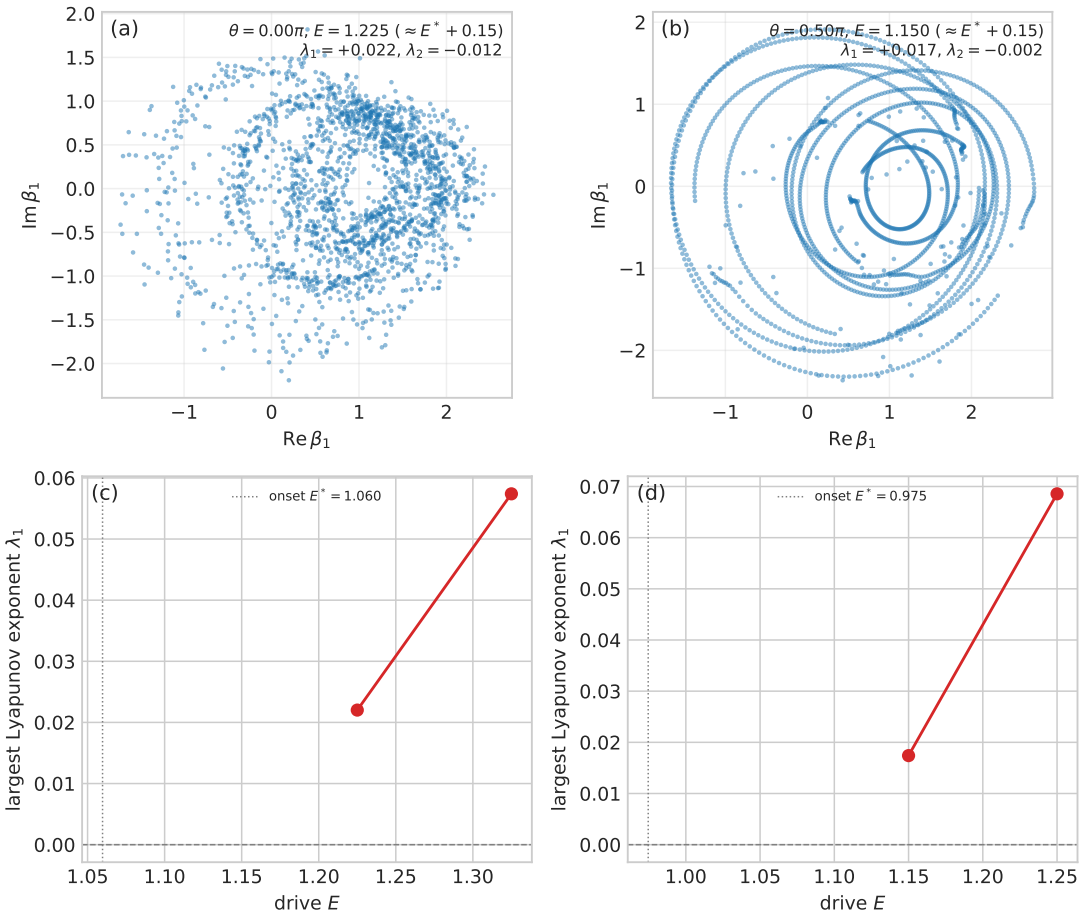



Figure 5: Observation of the Neimark–Sacker torus. (a),(b) Stroboscopic Poincaré sections (projected onto $\Re\beta_1, \Im\beta_1$) of the perturbed drive-locked orbit just above the onset at $\theta = 0$ and $\theta = \pi/2$, showing an invariant closed curve. (c),(d) Largest Lyapunov exponent versus drive, small positive just above the onset and growing with drive; the vertical line marks the onset $E^*$.

# 5 Discussion

Synthetic flux and nonlinear instability are the primary quantities of this work: Equation (5) defines the control coordinate and Equation (8) defines the dynamical observable used to certify hyperchaos order. Metrological information (Equation (12)) is evaluated only as a secondary geometric bound. Their interpretation requires each observable to be evaluated under explicit phase conventions and matched assumptions, which we do via the two statements of Sec. SI-5. The discussion below separates the *relation to prior work*, the *position among sensing benchmarks* (secondary), the *critical self-assessment* and the *bridge to experiment*.

## 5.1 Relation to prior work

The present study is a continuation at the level of model structure, not a parameter-for-parameter reproduction of Muthukumar *et al.* (the parameter-by-parameter difference is given in Table SI-1). The primary new question is whether the flux phase controls the *order* of a fully resolved hyperchaos. The answer is positive at the model level: the full spectrum

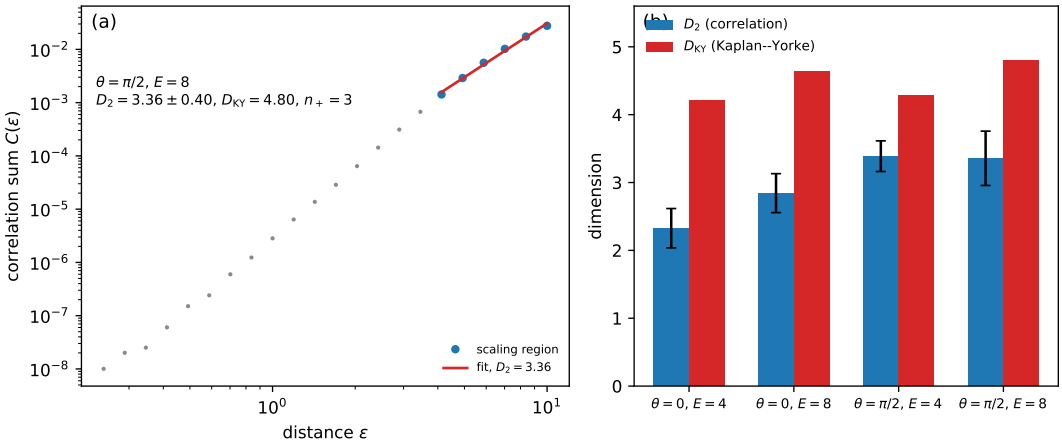



Figure 6: Correlation-dimension cross-check of the Kaplan–Yorke bound. (a) Theiler-corrected two-point correlation sum $C(\varepsilon)$ versus distance $\varepsilon$ at the most hyperchaotic point ($E = 8$, $\theta = \pi/2$); the scaling region and its fitted slope give the correlation dimension $D_2$. (b) Correlation dimension $D_2$ (with the scaling-region uncertainty) against the Kaplan–Yorke dimension $D_{\mathrm{KY}}$ at the four strong-coupling points; $D_2$ lies below $D_{\mathrm{KY}}$ at every point, as required by $D_2 \leq D_{\mathrm{KY}}$.

resolves up to four positive directions and the onset is localized to a Neimark–Sacker bifurcation. As a secondary geometric clarification, the matched Fisher calculation shows only a modest stable-orbit effect and no enhancement on the chaotic attractor. The scope is that of a nonlinear-dynamics study with a geometric measurement bound; a sharper quantum contribution would require a separate quantum validation.

## 5.2 Position among calibrated force-sensing benchmarks

Table 1 places the peak matched gain (1.32× over the flux-off sensor, 1.16× over the single-mode linear sensor) and the UQ median (1.039×, 1.019×) alongside four published optomechanical force-sensing benchmarks [16–19,33]: the present gain is of the same order of magnitude—real but modest, not an SQL-beating claim, and the flux-induced contribution is bounded by Lemma 1.

## 5.3 Critical self-assessment and limits of the present scope

Four limitations bound the strength of every claim above.

*(i) Dynamical scope and uniqueness.* The continued drive-locked orbit is the unstable seed of the route, not the attractor itself; the flux-induced modulation of the hyperchaos route is reproduced across three replicas and five drives (180/180 long-window runs completed), but the integer order varies across seeds near marginal exponents. Individual exponent magnitudes also vary by several percent between the two longest windows and are not interpreted as high-precision asymptotic estimates. The chaos/hyperchaos boundary is threshold-conventional at the $1 \times 10^{-3}$ cut (Table SI-8): the count is stable at four of twelve phases across all tested thresholds and threshold-sensitive at two marginal phases near the onset. A systematic basin-of-attraction analysis (or $\varepsilon$-machine reconstruction) that would elevate the three-seed audit to a uniqueness statement remains beyond the present scope; the reported order is therefore a robust feature of the sampled trajectories under the stated protocol, not yet a proven global property of the attractor. The computational cost of long-window integration ($2 \times 10^6$ steps per trajectory) limits the seed count;

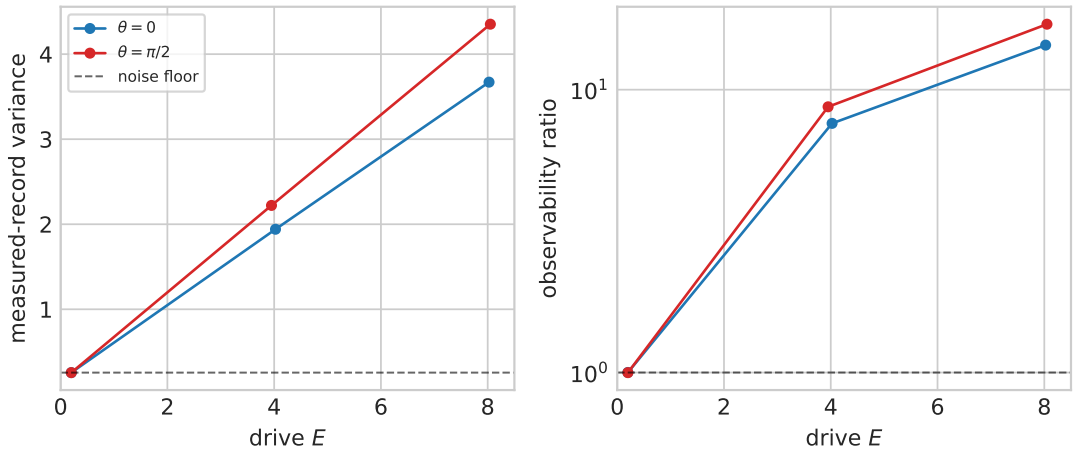



Figure 7: Noise-dependent observability of the chaotic signature. (a) Measured-record variance of $y = X_a + \nu_{\mathrm{det}}$ versus drive amplitude at $\theta = 0$ and $\theta = \pi/2$, with the noise-only floor (dashed line) set by the stable reference. (b) The observability ratio (measured variance divided by the noise floor); the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points sit 7.6–17.0× above the floor, while the stable reference sits at unity.

a larger ensemble would be required to elevate the audit to a uniqueness statement.

*(ii) Measurement scope and interpretation of the null result.* Matched force-sensing is the operational translation only at the weakly coupled, dissipation-dominated reference, where single-photon cooperativity $C = 0.08$ and mean cavity occupation $\langle|\alpha|^2\rangle \approx 0.032$ (sub-photon). Applied on the chaotic attractor, the same matched measurement resolves a flux-induced mean shift only at $E = 4$ and shows no enhancement: the bounded gain is a stable-orbit property, not a chaotic-transduction effect. The trajectory-separation argument [14, 27] is rejected here for the reason formalized in Lemma 1: noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max} t}$ as the deterministic displacement, so the matched signal-to-noise ratio cannot receive a multiplicative enhancement from trajectory separation alone. Physically, the flux phase reshapes the geometry of the unstable manifold (and therefore the hyperchaos order): a higher-dimensional unstable manifold offers more independent directions for perturbation growth, which in principle could enhance parameter estimation if each direction carried independent Fisher information. The matched-resource lemma shows, however, that under shared noise, power and observation time the additional directions simply partition the same total stretching budget rather than multiplying it, so the local stretching rates do not improve in a way that benefits the matched Fisher statistic. The null result on the chaotic attractor is therefore not a failure of the protocol but a direct consequence of the constraint: geometric complexity does not translate into metrological gain under the assumptions of Lemma 1. (The lemma bounds the classical Fisher statistic; it does not constrain quantum Fisher information, which may in principle exceed the classical bound.)

*(iii) Semi-classical validity.* All dynamical claims in the strong-coupling sector are statements about factorized mean-field trajectories. Truncated-Fock and truncated-Wigner checks confirm consistency of mean occupations at selected points (reference: 1.3 %; chaotic point: relative deviation $\leq 6.6 \times 10^{-6}$ at $N_{\mathrm{Fock}} = 60$; twelve Wigner trajectories yield $n_+ = 2$), yet these checks do not establish the accuracy of the factorization for every observable, nor do they bound quantum corrections to the Lyapunov spectrum or to the

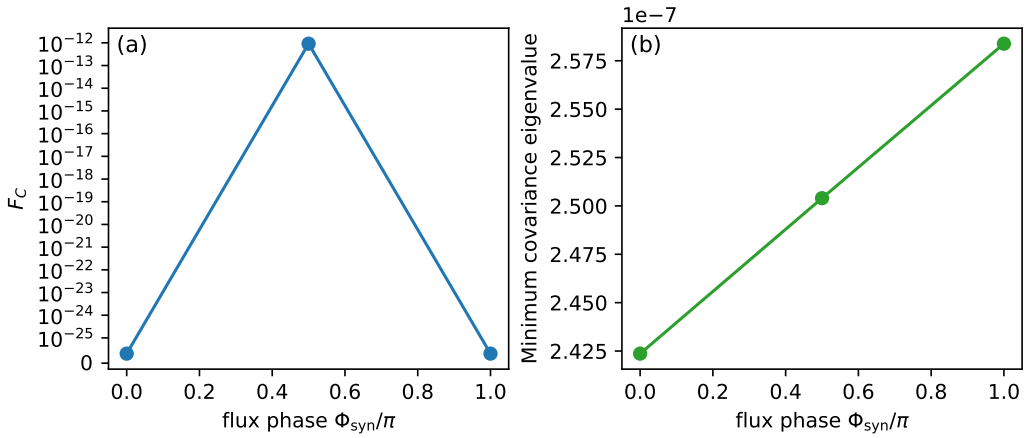



Figure 8: Covariance and measured-signal Fisher analysis for three synthetic-flux phases. (a) Classical Fisher information of the measured record. (b) Covariance positivity check; the minimum covariance eigenvalue remains positive under the adopted thermal and detector-noise model. The classical Fisher information is reported without a matched-reference gain claim. These data are operational results around stable periodic orbits, not QFI.

Fisher information. In the high-occupation regime ($E = 8$, mean mechanical occupations $\sim 20$) the dynamics are more classical than at the sub-photon reference, but a quantitative classical/quantum divergence bound for the full spectrum remains an open task (see Outlook).

*(iv) Experimental reach.* The strong-coupling regime requires $g_{1,2}/\omega_m \approx 0.3$ and $\gamma_m/\omega_m = 0.02$; these lie 2542×–2288× and 720× above the silicon-optomechanical-crystal anchors of [37, 38]; the feasibility-frontier scan places the hyperchaos onset near $g_1/\omega_m \approx 0.18$ (1525× the anchor) with no drive retuning lowering it (Table SI-11). The hyperchaos classification (and the secondary null metrological outcome) are therefore strictly model-level demonstrations. Any experimental claim would require ultrasonic characterization of the mechanical degeneracy $\omega_2$, a calibrated inter-resonator hopping $J_m$, and the emergence of a genuine gigahertz platform.

### 5.4 Bridge to experiment: mean-field boundary, route, and the 2542× calibration gap

The hyperchaos classification is a semiclassical statement about factorized mean-field dynamics; no quantum observable (Wigner contrast, quantum Fisher information) is computed or claimed in this sector. The dissipative trace is the state-independent constant $\mathrm{Tr}\, J = -1.04$. A truncated-Fock master equation for the driven cavity reproduces the reference mean occupation to 1.3 % and the chaotic occupation with a maximum relative deviation of $6.6 \times 10^{-6}$ at $N_{\mathrm{Fock}} = 60$ (Sec. SI-13). At the master-check point the mean occupations are $\langle a^\dagger a \rangle = 6.121$, $\langle b_1^\dagger b_1 \rangle = 20.417$ and $\langle b_2^\dagger b_2 \rangle = 23.538$. These values belong to that explicitly identified trajectory and do not calibrate the full strong-coupling map. Consistency with a more classical regime than the sub-photon reference is expected, yet it does not by itself justify a fully quantum treatment or establish the accuracy of factorization for every observable (see limitation (iii) above).

The transition is drive- and coupling-gated: the flux phase reshapes which directions are unstable and how many, but only after the drive and coupling place the system beyond the dissipation-dominated regime. The Lyapunov–Floquet consistency criterion and the

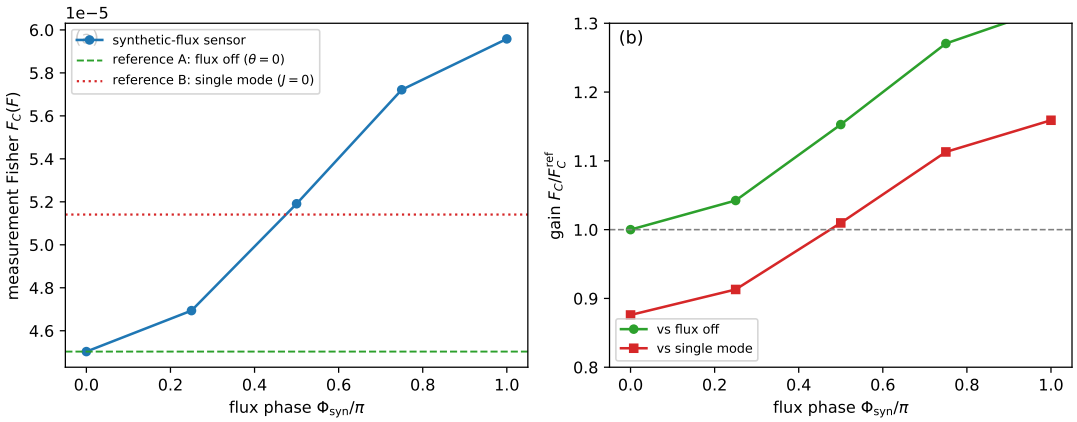



Figure 9: Matched measurement-Fisher reference for force sensing. (a) The classical Fisher information $F_C(F)$ for a weak external force on mechanical mode 1 versus the synthetic-flux phase, with the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$) overlaid; all configurations share the same drive, observation time, detector noise, baths, and estimator. (b) The gain relative to each reference; the flux phase yields at most a 1.32× (flux-off) and 1.16× (single-mode) enhancement, a modest order-unity effect.

matched-measurement Fisher lemma formalise this statement. The noise-resolved ensemble check provides a model-level observability test: the measured cavity quadrature stays 7.6–17.0× above the detector floor in the chaotic and hyperchaotic sectors, with the largest ratio at $\theta = \pi/2$, $E = 8$—the same phase and drive at which the hyperchaos order also peaks.

The drive-axis continuation of Figure 4 localizes the *route*: the first unstable direction emerges through a Neimark–Sacker torus bifurcation of the drive-locked orbit at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$), and the orbit family persists across the scanned drive range. Because the continued orbit is a periodic solution that becomes linearly unstable, its multipliers characterise the *onset* of the route rather than the attractor itself—the same distinction drawn in the Results section between the unstable seed and the attractor object. What remains is strictly experimental calibration: a measured $J_m$, an input power converted through pump frequency and external coupling, and fixed bath temperatures. A calibrated-device matched force-sensing comparison would close the gap quantified by the 2542× ratio reported in Table SI-14 and Table 1. The correlation-dimension estimate $D_2 \leq D_{KY}$ is a geometric characterization of the *sampled* attractor, not a proof of uniqueness (see limitation (i)).

## 6 Conclusions

Within a normalized six-dimensional dissipative optomechanical model we have established that a gauge-invariant synthetic-flux phase $\Phi_{syn}$ provides deterministic, reproducible control over the *order* of a drive- and coupling-gated hyperchaos transition. Up to four simultaneously positive Lyapunov exponents are resolved in the sampled strong-coupling map; the route is localized by a Neimark–Sacker bifurcation at $E^* \in 1.060$ and 0.975 and is certified by a three-layer numerical audit (Poincaré residual $< 10^{-8}$, dissipative volume balance $\sum \lambda_i = \mathrm{Tr}(J) = -1.04$, and truncated-Wigner robustness). This constitutes a model-level demonstration of synthetic-flux control of hyperchaos order beyond existing

Table 1: Matched-force-sensing gain in context. The present result is compared with optomechanical force-sensing and quantum-noise references that share the criterion of a homodyne/heterodyne measured record; the comparison is matched-resource (drive, observation time, bandwidth, baths, detector noise, estimator) and reports a numerical gain only, not a quantum or SQL-beating advantage.

| Reference | Reported gain (matched) | Operating regime |
|---|---|---|
| Teufel *et al.* (2011), pulsed sideband cooling | $\sim 2\times$ vs thermal bath | Sideband-resolved, $T = 10$ mK, mechanical $\omega_m/2\pi = 3.62$ MHz |
| Gavartin *et al.* (2012), optomechanical transducer | $\sim 3\times$ vs shot-noise floor | Photonic crystal nanobeam, room-temperature, $\omega_m/2\pi = 3.68$ GHz |
| Purdy *et al.* (2017), quantum-enhanced force sensing | up to $\sim 1.3\times$ beyond SQL | Pre-cooled membrane, $\omega_m/2\pi = 290$ MHz, $T = 10$ mK |
| Li *et al.* (2021), Kalman-filter tracking | $\sim 2\times$ over static estimator | Hot sideband-resolved cavity, $\omega_m/2\pi = 500$ MHz |
| Qvarfort *et al.* (2018), parametric amplification | $\sim 1.5\times$ over linear filter | Generic dispersive, no thermal assumption |
| **Present work (peak, weakly coupled reference)** | 1.32× flux-off, 1.16× single-mode | Normalized one-cavity/two-resonator, $\Phi_{\mathrm{syn}} \in [-\pi, \pi)$, readout cycle, $E = 0.2$, $T = 0.1$ |
| **Present work (UQ median, 60 MC + 20 000-bootstrap)** | 1.039× flux-off, 1.019× single-mode | Same with $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$ log-uniform, $n_{\mathrm{th}} \in [0.01, 1.0]$, $\pm 10\,\%$ drive |

single-mode benchmarks.

As a secondary geometric clarification, the same matched-resource force-sensing protocol yields only a modest enhancement on the stable reference orbit and a strictly null result on the chaotic attractor. The matched-Fisher lemma shows why: noise projected onto an unstable manifold is stretched by the identical local factor $e^{\lambda t}$ that amplifies the deterministic displacement; additional unstable directions merely partition a fixed stretching budget. Both the hyperchaos classification and the null metrological outcome remain strictly model-level: the explored sector sits 2542× beyond present silicon optomechanical-crystal anchors.

**Outlook.** Three concrete steps are required to elevate the present model-level control results: (i) ultrasonic characterization of the mechanical frequency degeneracy $\omega_2$ together with a calibrated measurement of the inter-resonator hopping $J_m$; (ii) emergence of a genuine gigahertz optomechanical platform that closes the 2542× coupling gap while preserving the synthetic-flux degree of freedom; (iii) systematic basin-of-attraction and $\varepsilon$-machine analysis that converts the three-seed audit into a uniqueness statement for the reported hyperchaos order, complemented by two-mode quantum-trajectory calculations that bound the classical/quantum divergence of the Lyapunov spectrum.

## Supplementary material

See the supplementary material for the physically anchored reference parameter set, the matched measurement-Fisher tables, the Lyapunov–Floquet consistency theorem, the periodic-covariance affine-fixed-point lemma, the matched-measurement Fisher lemma, and the representative convergence check of the strong-coupling spectrum.

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## Author contributions

SRMT and CT performed the numerical simulations and analyzed the data. PD and STK contributed to the theoretical framework and provided critical feedback on the results. SGNE conceived and supervised the project, directed the research, and wrote the final version of the manuscript. All authors reviewed and approved the final manuscript.

## Funding information

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## Competing interests

The authors declare no competing interests.

## Ethics

Not applicable. This is a purely theoretical and numerical study; no human or animal subjects, tissues, or clinical trials were involved.

## Data availability

The simulation and analysis scripts, environment specification, run manifests, checksums, source data, and figure-generation records supporting the findings of this study are available in the public repository https://github.com/NanaEngo/floquet-chaos-synthetic-flux. A Zenodo DOI will be minted and linked here upon publication. Every numerical claim in this article is traceable to the corresponding machine-readable data and source code in that repository.

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